

Adaptive Volumetric Parameterization of Simply Connected 3-Manifolds with Applications

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Abstract

Volumetric parameterization, the process of mapping a 3-manifold onto a simplified volumetric domain, is important for many tasks in computer graphics and imaging science. However, most prior volumetric parameterization approaches have only utilized standardized domains such as a solid ball regardless of the overall shape of the given 3-manifolds, which introduces significant geometric distortion and affects the subsequent shape processing and analysis tasks. To overcome this issue, in this work we propose a novel volumetric parameterization framework for simply connected 3-manifolds. Specifically, the proposed framework jointly controls local shape and mass distortions, while adapting the target domain during the optimization process. It enables three progressively more flexible target-domain settings for the parameterization: a prescribed solid ellipsoid, a volume-normalized adaptive ellipsoid with variable radii, and a sea-embedded free-boundary domain. For each setting, the parameterization algorithm consists of a 3D quasi-conformality shape update, a diffusion-based density-equalizing update, and a geometric correction procedure for removing element foldings, thereby allowing for volumetric parameterizations with different desired effects. Experimental results are presented to demonstrate the effectiveness of our proposed framework. Moreover, our framework can be easily applied to multiresolution and localized adaptive volumetric remeshing, volumetric registration, and volumetric morphing. Altogether, our work provides a new way for the representation, processing, and analysis of 3-manifolds.

1 Introduction

Shape parameterization, the process of mapping a complicated shape onto a simpler domain, has been widely studied in the areas of geometry processing, graphics, and imaging science [1, 2]. Over the past few decades, a large variety of works have been devoted to surface parameterization. In particular, conformal parameterization, which preserves angles and hence the local geometry [3], can be achieved by many approaches such as harmonic energy minimization [4–7], geometric flow [8–10], conformal welding [11], and circle patterns [12]. However, it is well-known that conformal parameterization may result in large area distortions. By contrast, authalic parameterization focuses on area preservation and can be achieved via various methods such as Lie advection [13], optimal mass transport [14–16], density-equalizing map [17–19], and stretch energy minimization [20]. Moreover, as a generalization of conformal maps, quasi-conformal maps [21, 22] have been widely applied to surface parameterization [23–28] to control the angular distortion and preserve bijectivity.

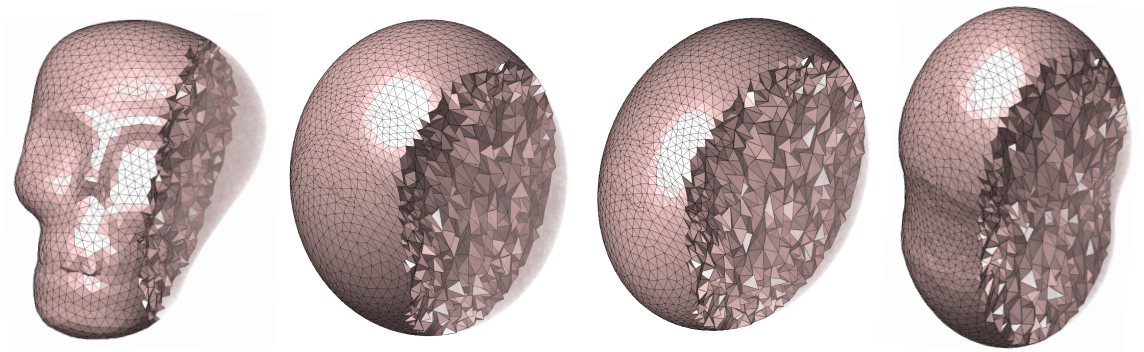


Figure 1: **An illustration of the proposed adaptive volumetric parameterization framework.** From left to right: An input 3-manifold skull model, a volume-preserving parameterization onto a prescribed ellipsoidal domain, a volume-preserving parameterization onto an optimized ellipsoidal domain, and a free-boundary volumetric parameterization based on a user-defined prescribed mass distribution.

Furthermore, other parameterization methods have also considered achieving a balance between angle and area distortions [29–33].

To represent 3D shapes in terms of not only their surface geometries but also their interior structures, volumetric meshes are commonly used. Analogous to surface parameterization, it is natural to utilize volumetric parameterization to map 3-manifolds onto simple canonical domains for shape analysis and modelling. However, compared with surface parameterization, volumetric parameterization is generally much more challenging due to the complex internal structure of those solid manifolds. In recent years, there has been an increasing interest in volumetric parameterization methods. For instance, Wang et al. [34] developed a method for computing volumetric harmonic maps. Naitsat et al. [35], Lee et al. [36], and Zhang et al. [37] proposed methods for computing n -dimensional quasi-conformal mappings. In [38], Su et al. proposed an algorithm for computing volumetric optimal mass transport mappings. In [39–41], Yueh et al. developed volumetric parameterization methods via stretch energy minimization, and Tan et al. [42] extended this framework into high-dimensional cases. More recently, Lyu et al. [43] developed a method for the volumetric parameterization of simply connected 3-manifolds onto a solid ball with controlled quasi-conformal and volumetric distortions. Other recent works on volumetric parameterization include close-to-conformal volumetric mappings [44, 45], and stretch-distortion energy [46], density-equalizing reference map [47], and isovolumetric energy minimization [48].

While many volumetric parameterization methods have been developed in the above-mentioned works, most of them have only considered the parameterization of volumetric meshes onto standardized domains such as a solid ball. For solid objects with more extreme geometries, such as highly elongated shapes, parameterizing them onto such domains may introduce significant geometric distortion, thereby posing challenges for subsequent shape processing and analysis tasks. To address this issue, in this work we propose a novel adaptive volumetric parameterization framework for simply connected 3-manifolds (see Fig. 1 for illustrations). Specifically, our proposed framework formulates the target domain as an intrinsic variable within the parameterization problem, while jointly controlling the local shape and mass distortions. Within this framework, we consider three

different target domains, each corresponding to different geometric constraints. For the prescribed ellipsoidal domain, the shape and radii of the target ellipsoid are fixed in advance, providing a common canonical domain for volumetric parameterization. For the volume-normalized adaptive ellipsoidal domain, the ellipsoid radii are allowed to vary while the target volume is preserved, thus enabling the ellipsoid radii to adapt to the input geometry and the prescribed mass distribution. For the sea-embedded free-boundary case, the ellipsoidal constraint is further relaxed, allowing the target boundary to deform within the sea. The effectiveness of our proposed methods has been verified through experimental results and practical applications. The contributions of this paper can be summarized as follows:

- We formulate the target-domain-adaptive volumetric parameterization as a joint problem in which both the mapping and the target domain are treated as optimization variables.
- We develop three volumetric parameterization models under different target-domain settings: a prescribed ellipsoid, a volume-normalized adaptive ellipsoid, and a sea-embedded free-boundary model.
- We develop numerical algorithms that combine quasi-conformal local shape distortion control, diffusion-based mass distortion control, and geometric correction for inverted tetrahedral elements.
- Our proposed methods can be utilized for various applications, including 3-manifold remeshing, registration, and morphing, demonstrating their efficacy and practicality.

The rest of the paper is organized as follows. In Section 2, we introduce the mathematical models for adaptive target-domain volumetric parameterization in detail. In Section 3, we describe the numerical details of our proposed algorithms. Experimental results are presented in Section 4 to assess the performance of our proposed methods. In Section 5, we introduce the applications of our methods to 3-manifold remeshing, registration, and morphing. We conclude our work and discuss possible future directions in Section 6.

2 Mathematical model

In our proposed framework, we develop a variational model to compute volumetric parameterizations of simply connected 3-manifolds. Unlike previous approaches, this model simultaneously controls both local shape and mass distortions while controlling fold-over throughout the parameterization process. In the following, we introduce the variational model and design a numerical algorithm to solve it.

2.1 Target domain adaptive variational model

Let $\mathcal{M}$ be a simply connected 3-manifold. In contrast to conventional volumetric parameterization methods that rely on a predefined target domain, our approach treats the target domain $\Omega \subset \mathbb{R}^3$ as an intrinsic variable of the model. Specifically, we aim to find a volumetric map $f : \mathcal{M} \rightarrow \Omega$ as well as the parameter domain Ω by minimizing the following energy formula

$$E(f, \Omega) = \alpha \int_{\mathcal{M}} |\log K_f|^2 + \beta \int_{\Omega} |\nabla \rho_{f, \Omega}|^2, \quad (1)$$

where $\alpha, \beta \geq 0$ balance the local shape distortion and the mass distortion. We denote the first and second terms by $E_{\text{shape}}(f)$ and $E_{\text{mass}}(f, \Omega)$, respectively. Here, K_f is the 3D quasi-conformal dilation of f , and $\rho_{f, \Omega}$ denotes the push-forward density of the prescribed mass measure under f . The two energy terms will be explained in detail in the following two subsections.

The proposed variational target-domain adaptive parameterization problem is given by

$$\min_{\Omega \in \mathcal{A}, f: \mathcal{M} \rightarrow \Omega} E(f, \Omega), \quad (2)$$

where $\mathcal{A}$ denotes a family of admissible target domains. The energy formulation also includes several special cases. By setting $\alpha = 1$ and $\beta = 0$, the model focuses on reducing local shape distortion. In contrast, when $\alpha = 0$ and $\beta = 1$, the model aims solely to equalize density. Furthermore, if $\alpha, \beta > 0$, the model can minimize the joint distortion.

The formulation in Eq. (1) includes both fixed-target and target-adaptive parameterization models. When the admissible family $\mathcal{A}$ contains only one prescribed domain, the problem reduces to a conventional fixed-target parameterization problem. When $\mathcal{A}$ contains multiple admissible domains, the target geometry becomes part of the optimization variables, allowing the parameter domain to adapt together with the volumetric map.

2.2 Local shape distortion via 3D quasi-conformality

We first introduce the 3D quasi-conformal dilation K utilized in our model. Let $f: \mathcal{M} \rightarrow \Omega$ be a volumetric map, and let $\mathbf{J}_f$ denote its associated Jacobian matrix. Following the 3D quasi-conformal representation in [49], we consider the modified polar decomposition

$$\mathbf{J}_f = RD, \quad (3)$$

where $R \in SO(3)$ is a rotation matrix with $\det(R) = 1$ and D is a symmetric matrix with eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$.

The 3D quasi-conformality is defined as $q = (\Lambda, \Theta)$, where $\Lambda = (\lambda_1, \lambda_2, \lambda_3)$ contains the eigenvalues of D and $\Theta = (\theta_x, \theta_y, \theta_z)$ contains the associated Euler angles. Here, Λ describes the principal stretches, while Θ determines their orientations. The mapping is conformal if and only if $\lambda_1 = \lambda_2 = \lambda_3$, or equivalently, $\lambda_1 = \lambda_3$. Moreover, with a suitable choice of the rotational component in the modified polar decomposition, the sign of λ_3 coincides with that of $\det(\mathbf{J}_f)$ [43].

To effectively quantify the local shape distortion of the mapping, like [43], we introduce the 3D quasi-conformal dilation as follows:

$$K(\lambda_1, \lambda_2, \lambda_3) = \begin{cases} \frac{\lambda_1}{\lambda_3}, & \text{if } \det(J_f) \neq 0, \\ \infty, & \text{if } \det(J_f) = 0, \end{cases} \quad (4)$$

where $\lambda_1 \geq \lambda_2 \geq \lambda_3$ are the eigenvalues of the matrix D . The dilation K quantifies shape distortion via its magnitude and encodes orientation through its sign. Specifically, when $K > 0$, the mapping is locally orientation-preserving and a value of K close to 1 indicates small local shape distortion.

2.3 Mass distortion via density equalization

We next introduce the density term. Let $(\mathcal{M}, \omega)$ be the input measured solid manifold, where ω denotes the prescribed mass measure on $\mathcal{M}$. For a volumetric map $f: \mathcal{M} \rightarrow \Omega$, the push-forward measure $f_*\omega$ describes the distribution of prescribed mass over the target domain Ω .

The volumetric mapping f is mass-preserving when the push-forward measure $f_*\omega$ is equivalent to the target Lebesgue measure dV up to a constant scaling factor:

$$f_*\omega = k_\Omega dV, \quad (5)$$

where $k_\Omega = \frac{\omega(\mathcal{M})}{\text{Vol}(\Omega)}$.

In general, there exists a density function $\rho_{f,\Omega} : \Omega \rightarrow (0, \infty)$ such that

$$f_*\omega = \rho_{f,\Omega} dV. \quad (6)$$

Therefore, Eq. (5) is equivalent to

$$\rho_{f,\Omega}(x) = k_\Omega \quad (7)$$

for $x \in \Omega$. At this density-equalized state, $E_{\text{mass}}(f, \Omega) = 0$.

2.4 Admissible target domains

The target-domain adaptive model in Eq. (1) depends on the choice of the admissible family $\mathcal{A}$. In this work, we consider three target-domain families: a prescribed ellipsoidal domain, a volume-normalized adaptive ellipsoidal domain, and a sea-embedded free-boundary domain.

2.4.1 Prescribed ellipsoidal domain

We first consider the case where the target domain is a prescribed solid ellipsoid. For given radii (a^0, b^0, c^0) , the solid ellipsoid is $\mathbb{E}_{a^0, b^0, c^0} = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{(a^0)^2} + \frac{y^2}{(b^0)^2} + \frac{z^2}{(c^0)^2} \leq 1 \right\}$. The corresponding admissible family is

$$\mathcal{A}_{\text{fix}} = \{\mathbb{E}_{a^0, b^0, c^0}\}. \quad (8)$$

Therefore, the target-domain adaptive problem reduces to

$$\min_{f: \mathcal{M} \rightarrow \mathbb{E}_{a^0, b^0, c^0}} E(f) = \alpha \int_{\mathcal{M}} |\log K_f|^2 + \beta \int_{\mathbb{E}_{a^0, b^0, c^0}} |\nabla \rho_f|^2. \quad (9)$$

Here, ρ_f denotes the density defined on the fixed target ellipsoid $\mathbb{E}_{a^0, b^0, c^0}$.

2.4.2 Volume-normalized adaptive ellipsoidal domain

The above method fixes the target shape throughout the optimization. However, a prescribed ellipsoid may not match the geometric proportions of the input 3-manifold, leading to unnecessary local shape distortion. To address this issue, we introduce an adaptive ellipsoidal domain model, which allows the ellipsoidal radii to vary.

For $a, b, c > 0$, let $\mathbb{E}_{a,b,c} = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1 \right\}$. Since $|\mathbb{E}_{a,b,c}| = \frac{4\pi}{3} abc$, fixing the product abc is equivalent to fixing the volume of the ellipsoidal target. We define the volume-normalized adaptive ellipsoidal domains by

$$\mathcal{A}_{\text{ellipsoidal}} = \{\mathbb{E}_{a,b,c} : abc = P_0, \quad a, b, c > 0\}, \quad (10)$$

where $P_0 > 0$ is determined by the initial solid ellipsoid.

The corresponding optimization problem is

$$\min_{\mathbb{E}_{a,b,c} \in \mathcal{A}_{\text{ellipsoidal}}, f: \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}} E(f, \mathbb{E}_{a,b,c}) = \alpha \int_{\mathcal{M}} |\log K_f|^2 + \beta \int_{\mathbb{E}_{a,b,c}} |\nabla \rho_{f, \mathbb{E}_{a,b,c}}|^2. \quad (11)$$

The constraint $abc = P_0$ removes the global scaling ambiguity and ensures that the model optimizes the aspect ratio of the target ellipsoid rather than its overall scale.

2.4.3 Free-boundary target domain

The adaptive ellipsoidal model still restricts the target domain to an ellipsoidal parametric family. In this section, we further relax this constraint and consider a free-boundary target model. In this free-boundary model, the target boundary is allowed to deform during the parameterization process.

In this case, the target domain is not restricted to be an ellipsoid. Instead, we consider a family of admissible target domains

$$\mathcal{A}_{\text{free}} = \{\Omega : \Omega \text{ is a simply connected admissible solid domain}\}. \quad (12)$$

The corresponding parameterization problem is

$$\min_{\Omega \in \mathcal{A}_{\text{free}}, f: \mathcal{M} \rightarrow \Omega} E(f, \Omega) = \alpha \int_{\mathcal{M}} |\log K_f|^2 + \beta \int_{\Omega} |\nabla \rho_{f, \Omega}|^2. \quad (13)$$

Compared with the ellipsoidal families, the free-boundary model relaxes the prescribed geometric constraint on the target domain and allows the boundary to adapt more freely to the input local shape and density distribution. In the numerical algorithm, the moving free boundary is handled by embedding Ω into a fixed outer computational domain with an auxiliary sea region. The details of this sea-embedded realization will be introduced in Section 3.3.1.

3 Proposed algorithm

In this section, we propose three numerical algorithms for the target domain adaptive models introduced in Section 2.

3.1 Prescribed ellipsoidal model

Consider a 3D solid manifold $\mathcal{M}$ in $\mathbb{R}^3$ with a closed surface boundary $\partial\mathcal{M}$ discretized as a tetrahedral mesh $(\mathcal{V}, \mathcal{E}, \mathcal{T})$, where $\mathcal{V}$ is the set of vertices, $\mathcal{E}$ is the set of edges, and $\mathcal{T}$ is the set of tetrahedral elements. Our goal is to develop an energy minimization algorithm for computing bijective volumetric parameterizations of $\mathcal{M}$ onto a prescribed ellipsoid balancing the local shape distortion (based on 3D quasi-conformality) and the mass distortion (based on a user-defined mass distribution).

3.1.1 Initial solid ellipsoid

Given a prescribed ellipsoid $\mathbb{E}_{a,b,c}$, we first compute an initial volumetric map $f^0 : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$. The initial map is computed in two steps. Firstly, we compute the boundary ellipsoidal parameterization

$f_\partial^0 : \partial\mathcal{M} \rightarrow \partial\mathbb{E}_{a,b,c}$ utilizing the ellipsoidal density-equalizing map (EDEM) method [19]. Then, the interior mapping is obtained by solving the Laplace equation

$$\begin{cases} \Delta f^0 = 0 & \text{in } \mathcal{M} \setminus \partial\mathcal{M}, \\ f^0(\partial\mathcal{M}) = \partial\mathbb{E}_{a,b,c}. \end{cases} \quad (14)$$

The resulting map is used as the initial map for the subsequent balanced optimization.

3.1.2 Local-shape descent

Let f^n be the mapping at the n -th iteration with 3D quasi-conformality $q^n = (\Lambda^n, \Theta^n)$. Specifically, for the eigenvalues $\Lambda^n = (\lambda_1^n, \lambda_2^n, \lambda_3^n)$, we first compute the residuals between the three eigenvalues as

$$R^n = \lambda_1^n - \lambda_2^n, \quad L^n = \lambda_2^n - \lambda_3^n, \quad (15)$$

where the residuals R^n and L^n are non-negative. When R^n and L^n vanish, all three eigenvalues become constant and the energy E_{shape} achieves the global minimum.

Then, we give the residual parameters

$$\tau^n = \frac{K^n - 1}{(K^n - 1) + \mathbf{C}}, \quad (16)$$

where $K^n = \frac{\lambda_1^n}{\lambda_3^n}$ is the 3D quasi-conformal dilation for f^n and $\mathbf{C} > 0$ is the weight constant.

Now, we utilize the residuals with their parameters to update the 3DQC q^n as follows. The eigenvalues Λ^{n+1} can be given by

$$\Lambda^{n+1} = (\lambda_1^{n+1}, \lambda_2^{n+1}, \lambda_3^{n+1}) = (\lambda_1^n - \tau^n R^n, \lambda_2^n, \lambda_3^n + \tau^n L^n), \quad (17)$$

where R^n, L^n are residuals and τ^n is the residual parameters. The updated 3DQC q^{n+1} is given by

$$q^{n+1} = (\Lambda^{n+1}, \Theta^{n+1}), \quad (18)$$

with $\Theta^{n+1} = \Theta^n$.

Remark 1. If $\lambda_1^n \geq \lambda_2^n \geq \lambda_3^n > 0$ and $0 \leq \tau^n \leq 1$, then the above residual update preserves the ordering of the dilation factors and satisfies

$$K^{n+1} = \frac{\lambda_1^{n+1}}{\lambda_3^{n+1}} \leq \frac{\lambda_1^n}{\lambda_3^n} = K^n. \quad (19)$$

Consequently, the local shape energy decreases during the iterative process:

$$E_{\text{shape}}^{n+1} \leq E_{\text{shape}}^n. \quad (20)$$

The algorithm is summarized in Algorithm 1.

Algorithm 1: Optimization of shape energy E_{shape}

Input: A simply connected 3-manifold $\mathcal{M}$, a weight constant $\mathbf{C}$, the elliptic radii a, b, c , the mapping $f^n : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$

Output: The updated 3DQC $q^{n+1} = (\Lambda^{n+1}, \Theta^{n+1})$.

- 1 Compute the 3DQC q^n and 3D quasi-conformal dilation K^n from f^n ;
 - 2 Compute the residuals R^n and L^n using Eq. (15);
 - 3 Compute the residual parameter τ^n using Eq. (16) ;
 - 4 Update q^{n+1} and K^{n+1} ;
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3.1.3 Density-equalizing descent

We next compute a density-equalizing descent direction for the mass term E_{mass} . The density-equalizing step is motivated by the diffusion equation [50]

$$\partial_t \rho = \Delta \rho, \quad (21)$$

which drives the density from high-density regions to low-density regions and therefore reduces spatial density variation. By diffusion theory and Fick's law, the velocity field is given by

$$\mathbf{v}(\mathbf{x}(t), t) = -\frac{\nabla \rho(\mathbf{x}(t), t)}{\rho(\mathbf{x}(t), t)}. \quad (22)$$

The tracer of $\mathbf{x}$ at time t can be computed by $\mathbf{x}(t) = \mathbf{x}(0) + \int_0^t \mathbf{v}(\mathbf{x}(\tau), \tau) d\tau$. As $t \rightarrow \infty$, the density induced by the diffusion process converges to a constant, yielding the density-equalizing map.

Remark 2. On a fixed ellipsoidal domain $\Omega = \mathbb{E}_{a,b,c}$ with Neumann boundary condition, the diffusion equation $\partial_t \rho = \Delta \rho$ dissipates the density Dirichlet energy:

$$\frac{d}{dt} \int_{\Omega} |\nabla \rho|^2 dV = -2 \int_{\Omega} |\Delta \rho|^2 dV \leq 0. \quad (23)$$

Thus, the diffusion process decreases E_{mass} on a fixed target domain.

Let $f^n : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$ be the n -th iteration mapping with density ρ^n . In the discrete case, the density on each tetrahedral element is defined by:

$$\rho_n^{\mathcal{T}} = \frac{m_T}{\text{Vol}(T)}, \quad (24)$$

where m_T is the prescribed mass assigned to T .

Then, we utilize the semidiscrete backward Euler method to solve the diffusion equation (21):

$$\frac{\rho_{n+1}^{\mathcal{V}} - \rho_n^{\mathcal{V}}}{\delta t} = \Delta_n \rho_{n+1}^{\mathcal{V}}, \quad (25)$$

where $\rho_n^{\mathcal{V}}$ is the n -th iterative vertex density, δt is the timestep, and Δ_n is the Laplace–Beltrami operator. On the tetrahedral mesh, the Laplace–Beltrami operator can be written as:

$$\Delta_n = -A_n^{-1} L_n. \quad (26)$$

Here A_n is a $|\mathcal{V}| \times |\mathcal{V}|$ diagonal matrix such that

$$(A_n)_i = \frac{1}{4} \sum_{\mathcal{T}_i \in \mathcal{NT}(\mathbf{x}_i)} \text{Vol}(\mathcal{T}_i), \quad (27)$$

and L^n is the volumetric cotangent Laplacian matrix defined by

$$(L_n)_{ij} = \begin{cases} h_{i,j} & \text{if } [\mathbf{x}_i, \mathbf{x}_j] \in \mathcal{E}, \\ -\sum_{k \neq i} h_{i,k} & \text{if } j = i, \\ 0 & \text{otherwise.} \end{cases} \quad (28)$$

Here, $h_{i,j} = -\sum_{s=1}^N l_s \frac{\cot(\theta_s)}{12}$ is the cotangent weight for volumetric Laplacian [34], where N is the number of tetrahedra sharing the edge $[\mathbf{x}_i, \mathbf{x}_j]$. θ_s is the dihedral angle opposite to the edge in the s -th incident tetrahedron, and l_s is the length of the corresponding opposite edge.

Therefore, Eq. (25) can be written as

$$\rho_{n+1}^\mathcal{V} = (A_n + \delta t L_n)^{-1} (A_n \rho_n^\mathcal{V}). \quad (29)$$

By solving this linear system, we can obtain the vertex density $\rho_n^\mathcal{V}$.

Next, we compute the density gradient on each tetrahedral element. For a tetrahedron $T = [\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k, \mathbf{x}_l]$, the vertex density is linearly interpolated via the Whitney 0-form:

$$\rho_n^\mathcal{V}(\mathbf{x}) = \rho_n^\mathcal{V}(\mathbf{x}_i) \Gamma_i^W(\mathbf{x}) + \rho_n^\mathcal{V}(\mathbf{x}_j) \Gamma_j^W(\mathbf{x}) + \rho_n^\mathcal{V}(\mathbf{x}_k) \Gamma_k^W(\mathbf{x}) + \rho_n^\mathcal{V}(\mathbf{x}_l) \Gamma_l^W(\mathbf{x}), \quad (30)$$

where $\Gamma_s^W(\mathbf{x})$ denotes the linear hat function associated with vertex $\mathbf{x}_s$. Since $\nabla \Gamma_i^W = -\frac{S_i}{6\text{Vol}(\mathcal{T})}$, the density gradient $\nabla \rho_{n+1}^\mathcal{T}(\mathcal{T})$ is given by

$$\nabla \rho_n^\mathcal{T}(\mathcal{T}) = -\frac{\rho_n^\mathcal{V}(\mathbf{x}_i) S_i + \rho_n^\mathcal{V}(\mathbf{x}_j) S_j + \rho_n^\mathcal{V}(\mathbf{x}_k) S_k + \rho_n^\mathcal{V}(\mathbf{x}_l) S_l}{6 \text{Vol}(\mathcal{T})}, \quad (31)$$

where the edges are denoted as $\vec{e}_{ij} = [\mathbf{x}_i, \mathbf{x}_j]$, and $S_i = (\vec{e}_{kl} \times \vec{e}_{kj})$, $S_j = (\vec{e}_{il} \times \vec{e}_{ik})$, $S_k = (\vec{e}_{ij} \times \vec{e}_{il})$, $S_l = (\vec{e}_{ik} \times \vec{e}_{ij})$. We can then easily convert $\nabla \rho_{n+1}^\mathcal{V}$ from $\nabla \rho_{n+1}^\mathcal{T}$ via the above-mentioned tetrahedral-to-vertex conversion matrix $M_\mathcal{V}^\mathcal{T}$:

$$\nabla \rho_n^\mathcal{V}(\mathbf{x}_i) = \frac{\sum_{T \in N_T(\mathbf{x}_i)} \text{Vol}(T) \nabla \rho_n^\mathcal{T}}{\sum_{T \in N_T(\mathbf{x}_i)} \text{Vol}(T)}. \quad (32)$$

The velocity field on each vertex is given as follows:

$$\mathbf{v}_n^\mathcal{V}(\mathbf{x}_i) = -\frac{\nabla \rho_n^\mathcal{V}(\mathbf{x}_i)}{\rho_n^\mathcal{V}(\mathbf{x}_i)}. \quad (33)$$

It is noteworthy that in the discrete case, the velocity on the boundary may not lie in the tangent space. To address this issue, we project the boundary velocity onto the tangent plane of the ellipsoidal boundary. Note that the equation of the ellipsoidal boundary is $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, and the

outward unit normal vector at $\mathbf{x}_{\text{bdy}}$ is $\mathbf{n}^\mathcal{V}(\mathbf{x}_{\text{bdy}}) = \frac{(\frac{x}{a^2}, \frac{y}{b^2}, \frac{z}{c^2})}{((\frac{x}{a^2})^2 + (\frac{y}{b^2})^2 + (\frac{z}{c^2})^2)^{\frac{1}{2}}}$. The projected boundary velocity can be computed via:

$$\tilde{\mathbf{v}}_n^\mathcal{V}(\mathbf{x}_{\text{bdy}}) = \mathbf{v}_n^\mathcal{V}(\mathbf{x}_{\text{bdy}}) - (\mathbf{v}_n^\mathcal{V}(\mathbf{x}_{\text{bdy}}) \cdot \mathbf{n}^\mathcal{V}(\mathbf{x}_{\text{bdy}})) \mathbf{n}^\mathcal{V}(\mathbf{x}_{\text{bdy}}), \quad (34)$$

where $\mathbf{x}_{\text{bdy}}$ denotes the boundary vertex. Ultimately, the velocity field $\bar{\mathbf{v}}_n^\mathcal{V}$ of diffusion flow is given by

$$\bar{\mathbf{v}}_n^\mathcal{V}(\mathbf{x}_i) = \begin{cases} \tilde{\mathbf{v}}_n^\mathcal{V}(\mathbf{x}_i) & \text{if } \mathbf{x}_i \in \partial\mathbb{E}_{a,b,c}, \\ \mathbf{v}_n^\mathcal{V}(\mathbf{x}_i) & \text{otherwise,} \end{cases} \quad (35)$$

Utilizing the above formulas, we can update the positions of all vertices based on Eq. (22):

$$\mathbf{x}_i(t_{n+1}) = \mathbf{x}_i(t_n) - \delta t \bar{\mathbf{v}}_n^\mathcal{V}(\mathbf{x}_i), \quad (36)$$

where δt is the time-step size. While the projection step can eliminate the normal components of the boundary velocity, there may still be some numerical errors that cause the boundary vertices to move out of the ellipsoidal boundary surface. To enforce the boundary constraint, we divide the coordinates of each boundary vertex $\mathbf{x}_{\text{bdy}}(t_{n+1}) = (x, y, z)$ by $\sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}}$. The diffusion-based method is summarized in Algorithm 2.

Algorithm 2: Descent direction for E_{mass}

Input: A simply connected 3-manifold $\mathcal{M}$, a user-defined prescribed mass distribution, radii (a, b, c) , and a solid ellipsoid map $f^n : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$.

Output: A solid ellipsoid map $f^{n+1} : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$.

- 1 Compute the n -th density $\rho_n^\mathcal{T}$ on $f^n(\mathcal{M})$;
 - 2 Let $\mathbf{x}_i(t_n) = f^n(v_i)$ for every vertex $v_i \in \mathcal{V}$;
 - 3 Compute A_n and L_n using Eq. (27) and Eq. (28);
 - 4 Obtain $\rho_{n+1}^\mathcal{V}$ by solving the diffusion equation (29);
 - 5 Compute the velocity field $\bar{\mathbf{v}}_{n+1}^\mathcal{V}$ using Eq. (33)–(35);
 - 6 Update the mapping $f^{n+1} = \mathbf{x}(t_{n+1})$ using Eq. (36);
-

3.1.4 Coupled update

After obtaining the descent directions of E_{shape} and E_{mass} , we now combine the local shape and density updates in the 3D quasi-conformality field. Let $f^n : \mathcal{M} \rightarrow E_{a,b,c}$ be the mapping at the n -th iteration with corresponding 3D quasi-conformality $q^n = (\Lambda^n, \Theta^n)$.

For the local shape term $E_{\text{shape}} = \int_{\mathcal{M}} |\log K|^2$, we utilize the residual method in Section 3.1.2 to compute the updated 3D quasi-conformality $\hat{q}^{n+1}$. The descent direction for the shape term is given by

$$dq_1^n = \hat{q}^{n+1} - q^n. \quad (37)$$

Then, we transform the descent direction of the mass energy E_{mass} from the physical spatial coordinates to the 3D quasi-conformality field. Specifically, using the diffusion process in Section 3.1.3, we can obtain the auxiliary updated mapping $\hat{f}^{n+1} : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$ with corresponding 3D quasi-conformality $\hat{q}^{n+1}$. Hence, dq_2^n can be obtained by

$$dq_2^n = \hat{q}^{n+1} - q^n. \quad (38)$$

Here, $\hat{f}^{n+1}$ is used to extract the density-induced variation in the quasi-conformality field.

The two updates are combined by

$$dq^n = \alpha dq_1^n + \beta dq_2^n, \quad (39)$$

and the updated 3D quasi-conformality field is given by

$$\bar{q}^{n+1} = q^n + \delta t dq^n, \quad (40)$$

where δt is the timestep. The corresponding updated 3D quasi-conformal dilation is evaluated as $\bar{K}^{n+1} = \frac{\bar{\lambda}_1^{n+1}}{\bar{\lambda}_3^{n+1}}$, where $\bar{\lambda}_1^{n+1} = \max(\bar{\Lambda}^{n+1})$ and $\bar{\lambda}_3^{n+1} = \min(\bar{\Lambda}^{n+1})$.

3.1.5 Geometric correction scheme

While the aforementioned procedure can efficiently update the 3D quasi-conformality field and optimize the energy, the bijectivity of the resulting volumetric mapping is not guaranteed. To resolve this issue, we propose a geometry correction scheme to rectify the overlaps via the 3D quasi-conformality field in this section.

Let $\mathbb{E}_{a,b,c}^n = f^n(\mathcal{M})$ be the solid ellipsoid at the n -th iteration. To eliminate boundary foldings on $\partial\mathbb{E}_{a,b,c}^n$, we apply the quasi-conformal truncation [19] to enforce $\|\mu\|_\infty < 1$, and map the corrected boundary back.

Having obtained a folding-free boundary, we then rectify the foldings in the tetrahedral mesh $\mathbb{E}_{a,b,c}^n$. Note that the 3D quasi-conformal dilation $\bar{K}^n$ encodes the orientation information. We denote the index set of folded tetrahedral elements at the current iteration as

$$\mathbb{I}_{\text{flip}}^n = \{i | \bar{K}_i^n \notin (0, +\infty)\}. \quad (41)$$

For each $i \in \mathbb{I}_{\text{flip}}^n$, we take absolute values of negative eigenvalues to correct orientation, and replace zero eigenvalues with a small $\vartheta > 0$. Subsequently, we reorder the modified eigenvalues in descending order to obtain the updated eigenvalues

$$\tilde{\Lambda}_i^n = (\tilde{a}_i^n, \tilde{b}_i^n, \tilde{c}_i^n), \quad i \in \mathbb{I}_{\text{flip}}^n. \quad (42)$$

The above steps fix folding but ignore geometric quality, which may result in extremely large updated dilation. To address this issue, we incorporate an additional geometric regularization step.

We set K_T as a prescribed distortion threshold. For the folded elements in $\mathbb{I}_{\text{flip}}^n$, we apply a localized affine transformation H_{K_T} to each individual component of the updated eigenvalues $\tilde{\Lambda}_i^n$:

$$H_{K_T}(x) = \begin{cases} l(x - e_{\text{mean}}) + e_{\text{mean}} & \text{if } \tilde{K}_i^n \geq K_T, \\ x & \text{otherwise,} \end{cases} \quad (43)$$

where $x \in \{\tilde{a}_i^n, \tilde{b}_i^n, \tilde{c}_i^n\}$. The centering parameter e_{mean} and the scaling factor l are formulated as:

$$e_{\text{mean}} = \frac{\tilde{a}_i^n + \tilde{c}_i^n}{2}, \quad \text{and} \quad l = \frac{(K_T - 1)e_{\text{mean}}}{(K_T + 1)(\tilde{a}_i^n - e_{\text{mean}})}. \quad (44)$$

It is clear that the transformation H_{K_T} modifies precisely those eigenvalues corresponding to extreme dilations. When $\tilde{K}_i^n \geq K_T$, the regularized dilation satisfies

$$\frac{H_{K_T}(\tilde{a}_i^n)}{H_{K_T}(\tilde{c}_i^n)} = \frac{\frac{2K_T e_{\text{mean}}}{K_T + 1}}{\frac{2e_{\text{mean}}}{K_T + 1}} = K_T. \quad (45)$$

Following the above regularization procedure, we can obtain the updated 3D quasi-conformality

$$q^n = (\Lambda^n, \Theta^n), \quad (46)$$

where $\bar{\Theta}^n = \Theta^n$ and $\Lambda_i^n = \begin{cases} \mathbb{T}(\bar{\Lambda}_i^n) & \text{if } i \in \mathbb{I}_{\text{flip}}^n, \\ \bar{\Lambda}_i^n & \text{otherwise.} \end{cases}$

To reconstruct the volumetric map from the corrected 3DQC field, we use the 3D quasi-conformal solver (3DQCS) [49]. Let $q = (\Lambda, \Theta)$ and let $W(\Theta)$ be the orthogonal matrix determined by Θ . Define

$$\mathcal{A}_q = W(\Theta) \text{diag} \left(\frac{\lambda_2 \lambda_3}{\lambda_1}, \frac{\lambda_1 \lambda_3}{\lambda_2}, \frac{\lambda_1 \lambda_2}{\lambda_3} \right) W(\Theta)^\top. \quad (47)$$

For $f = (u, v, w)$, 3DQCS solves

$$\nabla \cdot (\mathcal{A}_q \nabla u) = \nabla \cdot (\mathcal{A}_q \nabla v) = \nabla \cdot (\mathcal{A}_q \nabla w) = 0 \quad (48)$$

with the prescribed boundary condition. Its finite element discretization yields sparse linear systems for the three coordinate functions. We denote the reconstruction by

$$f^{n+1} = \text{3DQCS}(q^{n+1}). \quad (49)$$

For the overlaps adjacent to the boundary, we utilize the alternating freezing strategy. Specifically, the corrected interior elements are held fixed while the above correction scheme updates the remaining folding elements. The process is repeated until all tetrahedrons are orientation-preserving.

3.1.6 Summary

The proposed algorithm is summarized in Algorithm 3. In practice, we set the timestep $\delta t = 0.1$, the stopping parameter $\epsilon = 10^{-2}$, and the maximum number of iteration $n_{\max} = 100$.

Algorithm 3: Prescribed ellipsoidal volumetric map

- Input:** A simply connected 3-manifold $\mathcal{M}$, a user-defined mass distribution, weighting parameters α, β , the stopping parameter ϵ , the maximum number of iterations allowed $n_{\max}$, and the threshold K_T .
- Output:** A bijective solid ellipsoid map $f : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$.
- 1 Compute the initial solid ellipsoid map $f^0 : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$;
 - 2 Compute the initial density $\rho_0^\mathcal{T}$, initial 3D quasi-conformality q_0 , and the initial 3D quasi-conformal dilation K^0 on $f_0(\mathcal{M})$;
 - 3 Set $n = 0$;
 - 4 **repeat**
 - 5 Use the residual method (Algorithm 1) to obtain the descent direction dq_1^n for E_{shape} ;
 - 6 Use the density-equalizing method (Algorithm 2) to obtain the descent direction dq_2^n for E_{mass} ;
 - 7 Compute the updated $\tilde{q}^{n+1}$ and $\tilde{K}^{n+1}$ using the prescribed weights α, β and Eq. (40);
 - 8 Apply the correction scheme in Section 3.1.5 with the threshold K_T to obtain the updated map f^{n+1} ;
 - 9 Update $\rho_{n+1}^\mathcal{T}$ based on f^{n+1} ;
 - 10 Set $n = n + 1$;
 - 11 **until** $|f^{n+1} - f^n| < \epsilon$ **or** $n \geq n_{\max}$;
 - 12 **return** $f = f^N$, where N is the total number of iterations;
-

3.2 Volume-normalized adaptive ellipsoidal model

According to Section 2.4.2, we next introduce the volume-normalized adaptive ellipsoidal model, which allows the ellipsoidal target region to adaptively adjust during the optimization process.

Based on Eq. (10), the radii in this model must satisfy $abc = P_0$, where $P_0 = a_0 b_0 c_0$ is determined by the initial target ellipsoid. Therefore, the target-shape update changes the aspect ratio of the ellipsoid while keeping its volume factor fixed.

3.2.1 Decoupling the combined energy

To solve the volume-normalized adaptive ellipsoidal model (11), we decouple the optimization into two subproblems, where the mapping f and the radii (a, b, c) are optimized independently:

- Subproblem 1 (Mapping Update): With the ellipsoidal radii (a, b, c) fixed, we update the volumetric mapping f by minimizing:

$$E_1(f) = \alpha \int_{\mathcal{M}} |\log K|^2 + \beta \int_{\mathbb{E}_{a,b,c}} |\nabla \rho|^2. \quad (50)$$

- Subproblem 2 (Radii Optimization): we perform a local ellipsoidal radii search under the volume constraint:

$$E_2(a, b, c) = \alpha \int_{\mathcal{M}} |\log K|^2 + \beta \int_{\mathbb{E}_{a,b,c}} |\nabla \rho|^2, \quad (51)$$

with the volume-normalization constraint $abc = P_0$.

In our numerical algorithm, these two subproblems are solved at an asymmetric iteration frequency. Specifically, subproblem 1 is executed multiple times to fully deform the mesh vertices, followed by a single execution of subproblem 2 to adjust the target domain. For fixed radii, the mapping update is exactly the prescribed ellipsoidal procedure described in Section 3.1. Specifically, we compute the local shape and density updates, combine them in the 3D quasi-conformality field, and apply the geometric correction before reconstructing the updated map. The target-shape update is described in the next subsection.

3.2.2 Radii optimization

For Subproblem E_2 , we optimize the ellipsoidal radii under a volume-normalization constraint. Let $P_0 = a_0 b_0 c_0$ be the prescribed volume factor and $\gamma = P_0^{1/3}$. We parameterize the ellipsoidal radii by two log-radii variables (u, v) :

$$a = \gamma e^u, \quad b = \gamma e^v, \quad c = \gamma e^{-u-v}, \quad (52)$$

which satisfies $abc = \gamma^3 e^u e^v e^{-u-v} = P_0$. Hence, the target-shape update is reduced to a two-dimensional optimization in the (u, v) -plane, and only the aspect ratio of the ellipsoid is changed.

For the volumetric map $f^n : \mathcal{M} \rightarrow \mathbb{E}_{a^n, b^n, c^n}$, log-radii variables are

$$u^n = \log \frac{a^n}{\gamma}, \quad v^n = \log \frac{b^n}{\gamma}. \quad (53)$$

With the search step size δ_r , we build a 3×3 stencil in the (u, v) -plane:

$$(\bar{u}, \bar{v}) = (u^n + \ell_u \delta_r, v^n + \ell_v \delta_r), \quad \ell_u, \ell_v \in \{-1, 0, 1\}. \quad (54)$$

For each $(\bar{u}, \bar{v})$, the corresponding radii are $(\bar{a}, \bar{b}, \bar{c}) = (\gamma e^{\bar{u}}, \gamma e^{\bar{v}}, \gamma e^{-\bar{u}-\bar{v}})$ such that $\bar{a}\bar{b}\bar{c} = P_0$.

According to the updated radii $(\bar{a}, \bar{b}, \bar{c})$, we can compute the candidate ellipsoidal boundary condition $(x_1, x_2, x_3) \mapsto \left(\frac{\bar{a}}{a^n} x_1, \frac{\bar{b}}{b^n} x_2, \frac{\bar{c}}{c^n} x_3\right)$. Then we obtain the candidate volumetric mappings $f_{\bar{u}, \bar{v}}$ via 3DQCS with associated 3D quasi-conformality q^n .

For each candidate $(\bar{u}, \bar{v})$, we evaluate the full energy $E(\bar{u}, \bar{v}) = E(f_{\bar{u}, \bar{v}}, \mathbb{E}_{\bar{a}, \bar{b}, \bar{c}})$. Then we choose the smallest one

$$(u^{n+1}, v^{n+1}) = \underset{(\bar{u}, \bar{v}) \in \mathcal{S}(u^n, v^n)}{\operatorname{argmin}} E(\bar{u}, \bar{v}), \quad (55)$$

where $\mathcal{S}(u^n, v^n)$ denotes the above 3×3 set. The updated radii are $(a^{n+1}, b^{n+1}, c^{n+1}) = (\gamma e^{u^{n+1}}, \gamma e^{v^{n+1}}, \gamma e^{-u^{n+1}-v^{n+1}})$ such that $a^{n+1}b^{n+1}c^{n+1} = P_0$.

Finally, we apply a uniform normalization to eliminate numerical errors and enforce the target volume P_0 . Let $s^{n+1} = \left(\frac{a^{n+1}b^{n+1}c^{n+1}}{P_0}\right)^{1/3}$. The mapping is updated as

$$(a^{n+1}, b^{n+1}, c^{n+1}, f^{n+1}) \leftarrow \left(\frac{a^{n+1}}{s^{n+1}}, \frac{b^{n+1}}{s^{n+1}}, \frac{c^{n+1}}{s^{n+1}}, \frac{1}{s^{n+1}} f^{n+1}\right). \quad (56)$$

This normalization uniformly rescales the map and radii, preserving the aspect ratio while satisfying the volume constraint.

The algorithm for shape optimization is summarized in Algorithm 4.

3.2.3 Summary

Combining the two subproblems yields the volume-normalized adaptive ellipsoidal parameterization algorithm. Starting from an initial map with radii (a_0, b_0, c_0) , we apply the mapping-update procedure for H iterations on the current ellipsoid. The geometric correction is employed at each iteration. We then perform one target-shape update under the fixed-volume constraint. At the m -th target-shape update, the log-radii search step is set to $\delta_r^{(m)} = 0.9^{m-1} \delta_r$. The iterative process is repeated until $\|f^{n+1} - f^n\| < \epsilon$ or $n = n_{\max}$, as summarized in Algorithm 5. In practice, the initial radii can be selected arbitrarily, the timestep is 0.1, the initial radius step sizes are $\delta_r = 0.1$, the number of iterations for each set of fixed radii is $H = 5$, the error threshold is $\epsilon = 10^{-3}$, and the maximum number of iterations is $n_{\max} = 100$.

3.3 Sea-embedded free-boundary model

Both the prescribed and volume-normalized adaptive models restrict the target domain to a strict solid ellipsoid. To further relax this target-domain constraint, we introduce a sea-embedded free-boundary target model. By embedding the target domain Ω inside an auxiliary sea domain, Ω can deform freely, which simultaneously prevents unbounded computational expansion.

Algorithm 4: Shape update of the ellipsoidal domain

Input: Current volumetric map $f^n : \mathcal{M} \rightarrow \mathbb{E}_{a^n, b^n, c^n}$, current radii (a^n, b^n, c^n) , prescribed mass distribution, current 3D quasi-conformality field q^n , step size δ_r .

Output: Updated map f^{n+1} and updated radii $(a^{n+1}, b^{n+1}, c^{n+1})$ satisfying $a^{n+1}b^{n+1}c^{n+1} = P_0$.

- 1 Set $\gamma = P_0^{1/3}$;
 - 2 Compute the current log-radii variables $u^n = \log \frac{a^n}{\gamma}, v^n = \log \frac{b^n}{\gamma}$;
 - 3 Generate the local candidate set $\mathcal{S}(u^n, v^n) = \{(u^n + \ell_u \delta_r, v^n + \ell_v \delta_r) : \ell_u, \ell_v \in \{-1, 0, 1\}\}$;
 - 4 **for** $(\bar{u}, \bar{v}) \in \mathcal{S}(u^n, v^n)$ **do**
 - 5 Compute $(\bar{a}, \bar{b}, \bar{c}) = (\gamma e^{\bar{u}}, \gamma e^{\bar{v}}, \gamma e^{-\bar{u}-\bar{v}})$;
 - 6 Rescale the boundary image from $\partial \mathbb{E}_{a^n, b^n, c^n}$ to $\partial \mathbb{E}_{\bar{a}, \bar{b}, \bar{c}}$;
 - 7 Reconstruct the candidate map $f_{\bar{u}, \bar{v}}$ by 3DQCS using q^n and the updated boundary condition;
 - 8 Evaluate energies $E(\bar{u}, \bar{v}) = E(f_{\bar{u}, \bar{v}}, \mathbb{E}_{\bar{a}, \bar{b}, \bar{c}})$;
 - 9 Choose $(u^{n+1}, v^{n+1}) = \arg \min_{(\bar{u}, \bar{v}) \in \mathcal{S}(u^n, v^n)} E(\bar{u}, \bar{v})$;
 - 10 Set $(a^{n+1}, b^{n+1}, c^{n+1}) = (\gamma e^{u^{n+1}}, \gamma e^{v^{n+1}}, \gamma e^{-u^{n+1}-v^{n+1}})$;
 - 11 Set $f^{n+1} = f_{u^{n+1}, v^{n+1}}$;
 - 12 Apply a final uniform normalization to $(a^{n+1}, b^{n+1}, c^{n+1}, f^{n+1})$ to remove numerical error;
 - 13 **return** f^{n+1} and $(a^{n+1}, b^{n+1}, c^{n+1})$;
-

3.3.1 Construction of the sea-embedded domain

Let $f : \mathcal{M} \rightarrow \Omega$ be the initial map, where $\Omega = \mathbb{E}_{a, b, c}$ denotes a solid ellipsoidal tetrahedral mesh. Specifically, we denote the set of boundary triangular faces and the corresponding boundary vertices as $\mathcal{F}_{\partial\Omega}$ and $\mathcal{V}_{\partial\Omega}$, respectively.

The auxiliary sea region is constructed by generating several concentric layers outward along the boundary surface $\mathcal{F}_{\partial\Omega}$. Let $\bar{l}$ denote the mean boundary edge length in Ω . The thickness of the layer is defined as

$$h = \kappa \bar{l}, \quad (57)$$

where κ is a positive constant. For each boundary vertex $v_i \in \mathcal{V}_{\partial}$, we then generate the corresponding sea vertices by radial scaling:

$$w_{k,i} = s_k v_i, \quad k = 1, \dots, L, \quad (58)$$

where $s_k = 1 + kh$ and L is the number of layers.

Each boundary triangle $f = (i, j, \ell) \in \mathcal{F}_{\partial\Omega^0}$ forms a triangular prism between adjacent layers. We subdivide each prism into three tetrahedrons across all boundary faces, and all sea layers yield a tetrahedral shell mesh Ω_{sea} . The sea-embedded domain is then defined as

$$\bar{\Omega} = \mathbb{E}_{\bar{a}, \bar{b}, \bar{c}} = \Omega \cup \Omega_{\text{sea}}, \quad (59)$$

with $\bar{a} = s_L a$, $\bar{b} = s_L b$, and $\bar{c} = s_L c$.

Within $\bar{\Omega}$, the original boundary $\partial\Omega$ becomes an interior interface, while the outermost sea layer forms the new boundary $\partial\bar{\Omega}$. The sea serves as a volumetric buffer region for density diffusion and improves the stability of the free-boundary optimization process.

The algorithm for the construction of the sea is summarized in Algorithm 6.

Algorithm 5: Volume-normalized adaptive ellipsoidal map

Input: A simply connected 3-manifold $\mathcal{M}$, a user-defined mass distribution, weight parameters α, β , the threshold K_T , the initial elliptic radii (a_0, b_0, c_0) , the step size δt , the radius step sizes δ_r , the number of iterations for each set of fixed radii H , the stopping parameter ϵ , and the maximum number of iterations allowed $n_{\max}$.

Output: A volumetric ellipsoidal parameterization $f : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$ with adapted radii (a, b, c) .

- 1 Set $(a, b, c) = (a_0, b_0, c_0)$;
- 2 Compute the initial solid ellipsoid map $f^0 : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$;
- 3 Compute the initial density $\rho_0^\mathcal{T}$, initial 3D quasi-conformality q_0 , and the initial 3D quasi-conformal dilation K^0 on $f_0(\mathcal{M})$;
- 4 Set $n = 0$;
- 5 **repeat**
- 6 Apply the residual method (Algorithm 1) and density-equalizing method (Algorithm 2) to compute dq^n ;
- 7 Update the 3D quasi-conformality $\bar{q}^{n+1} = q^n + \delta t dq^n$;
- 8 Apply the correction scheme in Section 3.1.5 to obtain the updated map f^{n+1} ;
- 9 Update $\rho_{n+1}^\mathcal{T}$ based on f^{n+1} ;
- 10 **if** $n = Hm$ for some positive integer m **then**
- 11 Apply the shape update method (Algorithm 4) with the radius step sizes $0.9^m \delta_r$ to obtain the updated map f^{n+1} and updated radii (a, b, c) ;
- 12 Update $\rho_{n+1}^\mathcal{T}$, q^{n+1} , and K^{n+1} on the updated ellipsoid $\mathbb{E}_{a,b,c}$;
- 13 Set $n = n + 1$;
- 14 **until** $|f^{n+1} - f^n| < \epsilon$ **or** $n \geq n_{\max}$;
- 15 **return** The result map is $f : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$, where (a, b, c) are the optimal elliptic radii;

3.3.2 Free-boundary update

Next, we update the parametrization on this sea-embedded domain $\bar{\Omega} = \Omega^n \cup \Omega_{\text{sea}}^n$. Specifically, the density within the auxiliary sea Ω_{sea}^n is set to the mean value of the initial density.

The update on the original domain follows the same coupled strategy as in the prescribed ellipsoidal model. We compute the local shape and density descents and combine them to deform the mapping. Note that we omit the projection step for the vertices on the original boundary. Therefore, the original domain is treated as an interior part and can move freely in $\bar{\Omega}^n$. For the auxiliary sea domain, we focus solely on the density-driven deformation and omit the local shape distortion. Moreover, we apply a projection step for the vertices on the outside boundary of the sea-embedded domain to project them back to the fixed outer ellipsoid $\partial\bar{\Omega}$.

Remark 3. *The auxiliary sea region is not included in the local shape distortion term. At the beginning of each iteration, its density is reset to the mean density of the current original domain. Hence, the sea does not carry an independent mass distribution or generate an internal density gradient. It serves as a constant-density buffer for the free-boundary update, while interacting with the original domain through the interface during the diffusion step.*

Algorithm 6: Construction of sea

Input: A tetrahedral mesh of the ellipsoid Ω , number of layers L , layer ratio κ .

Output: An extended tetrahedral mesh $\bar{\Omega}$ with a volumetric sea.

- 1 Extract the boundary triangulation $\mathcal{F}_{\partial\Omega}$;
 - 2 Compute the mean boundary edge length $\bar{l}$ and set $h = \kappa\bar{l}$;
 - 3 Generate L concentric ellipsoidal layers by radial scaling;
 - 4 Connect adjacent layers to form triangular prisms;
 - 5 Subdivide each prism into tetrahedra using a standard prism decomposition;
 - 6 Assemble the sea mesh and merge it with the original tetrahedral mesh;
 - 7 **return** *The sea-embedded domain $\bar{\Omega}$;*
-

3.3.3 Summary

In summary, the sea-embedded free-boundary model updates the original target domain together with an auxiliary sea region. The original tetrahedra follow the coupled shape-density update, whereas the sea tetrahedra are used only for the density-driven update. The fixed outer ellipsoid prevents uncontrolled expansion, while the sea region provides auxiliary degrees of freedom without introducing an additional energy contribution. The complete sea-embedded free-boundary method is summarized in Algorithm 7.

Algorithm 7: Sea-embedded free-boundary volumetric parameterization

Input: A simply connected 3-manifold $\mathcal{M}$, a prescribed mass distribution, weight parameters α, β , initial ellipsoid radii (a_0, b_0, c_0) , sea parameters L, κ , timestep δt , stopping parameter ϵ , and maximum iteration number $n_{\max}$.

Output: A free-boundary volumetric parameterization $f : \mathcal{M} \rightarrow \mathbb{R}^3$.

- 1 Compute an initial ellipsoidal map $f^0 : \mathcal{M} \rightarrow \Omega^0 = \mathbb{E}_{a_0, b_0, c_0}$;
 - 2 Apply the construction scheme in Section 3.3.1 to obtain sea-embedded domain $\bar{\Omega} = \Omega^0 \cup \Omega_{\text{sea}}^0$;
 - 3 Initialize the density on $\bar{\Omega}$;
 - 4 Set $n = 0$;
 - 5 **repeat**
 - 6 Compute the diffusion-driven update on the $\bar{\Omega}$;
 - 7 Compute the local shape-driven update on the original domain Ω^n ;
 - 8 Combine the two directions to obtain the updated map on $\bar{\Omega}$;
 - 9 Apply the geometric correction scheme to $\bar{\Omega}$;
 - 10 Set $n = n + 1$;
 - 11 **until** $|f^{n+1} - f^n| < \epsilon$ **or** $n \geq n_{\max}$;
 - 12 Remove the auxiliary sea;
 - 13 **return** $f = f^N$, where N is the total number of iterations;
-

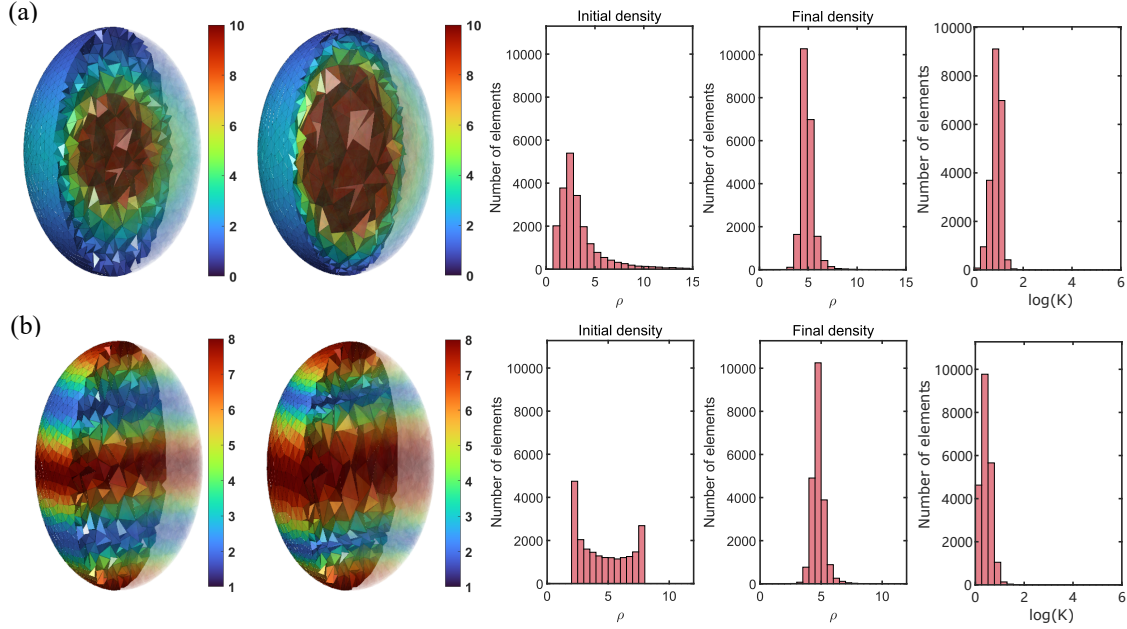


Figure 2: **Ellipsoidal mass-preserving quasi-conformal maps of solid ellipsoids.** Each row shows one example. (a) An example with a continuous input density that varies with the distance to the origin. (b) An example with a continuous input density that varies with the z -coordinate. Left to right: The initial solid ellipsoid color-coded with the initial density, the final result color-coded with the initial density, the histogram of the initial density, the histogram of the final density, and the histogram of $\log(K)$.

4 Experimental results

In this section, we present experimental results to demonstrate the effectiveness of the proposed volumetric parameterization framework. The proposed algorithms are implemented in MATLAB R2021a. All experiments are conducted on a Windows computer with an Intel(R) Core(TM) i9-12900 2.40 GHz processor and 32 GB of memory. All 3-manifolds are discretized in the form of tetrahedral meshes.

4.1 Prescribed ellipsoidal model

We first test our proposed prescribed ellipsoidal method by mapping some solid ellipsoids with different density distributions. In the examples shown in Fig. 2, we define different mass distributions on the solid ellipsoid to obtain different initial densities. It can be observed that the initial densities are highly non-uniform in both examples. We then apply the proposed method and obtain the prescribed ellipsoidal mapping results as shown in the figure. Compared with the initial maps, the results enlarge the high-density regions while significantly shrinking the low-density regions. From the density histograms, we can see that the density is effectively equalized. Moreover, the histograms of $\log(K)$ show that the results effectively control local shape.

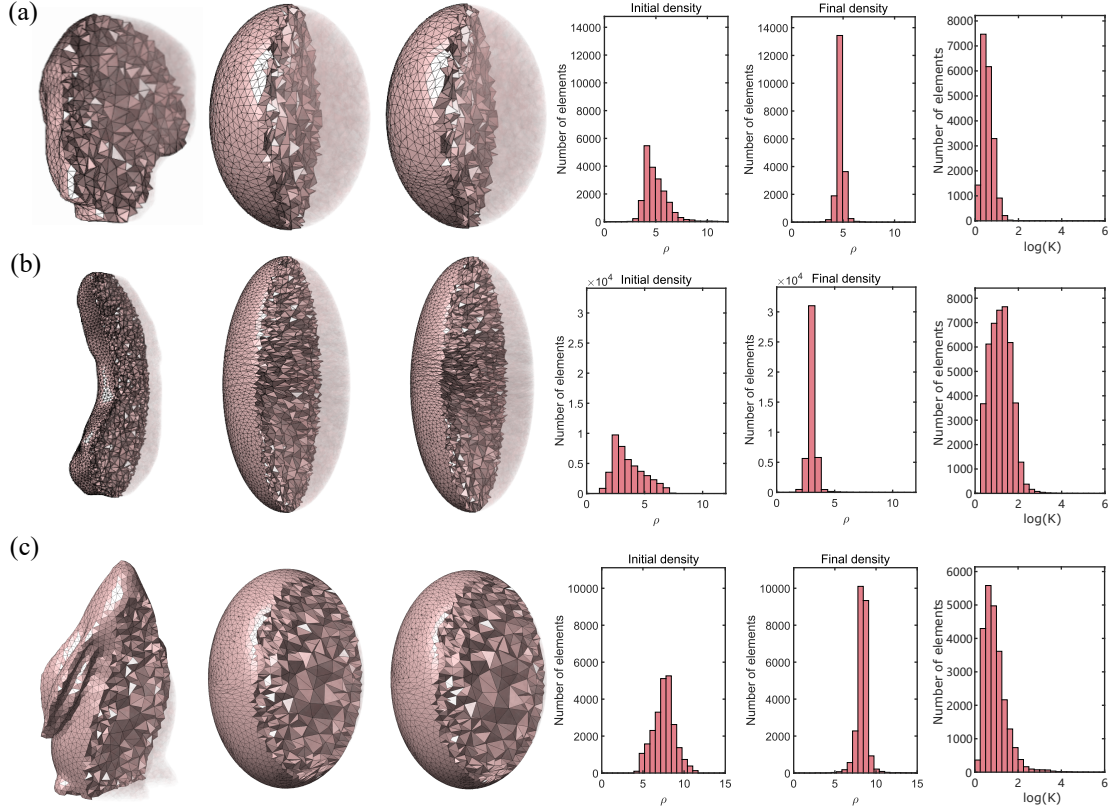


Figure 3: **Ellipsoidal volumetric parameterization of simply connected 3-manifolds obtained by our method.** Each row shows one example. (a) The Igea model. (b) The Hippocampus model. (c) The Rabbit model. Left to right: The original 3-manifolds, the initial solid ellipsoid, the final result, the histogram of the initial density, the histogram of the final density, and the histogram of $\log(K)$.

Then we consider the prescribed ellipsoidal mass-preserving quasi-conformal parameterizations for general simply connected manifolds. In Fig. 3, we prescribe different mass distributions on the input tetrahedral meshes. The initial density histograms in Fig. 3 show that the density distortions are large in the initial ellipsoidal parameterization for all examples. Then we apply the proposed method and obtain the mapping results. The final density histograms in Fig. 3 are highly concentrated, indicating that the prescribed mass is more uniformly distributed over the target ellipsoids. Furthermore, the local shape distortion histograms in Fig. 3 concentrate near zero, which shows that the local shape distortions are controlled effectively.

For a more quantitative analysis of our prescribed ellipsoidal volumetric parameterization method, we record the radii of the target ellipsoidal domain, variance of the initial density, variance of the final density, mean and standard deviations of the 3D quasi-conformal dilation K , and number of overlaps for mapping different models (see Table 1). Compared with the initial density, the variance

Table 1: **The performance of our prescribed ellipsoidal model.** For each example, we record the number of tetrahedral elements, the radii (a, b, c) , the variance of the normalized initial density $\tilde{\rho}_0 = \frac{\rho_0}{\text{Mean}(\rho_0)}$, the normalized final density $\tilde{\rho}_1 = \frac{\rho_1}{\text{Mean}(\rho_1)}$, the mean and standard deviations of the 3D quasi-conformal dilation K .

Example	# Elements	Radii (a, b, c)	$\text{Var}(\tilde{\rho}_0)$	$\text{Var}(\tilde{\rho}_1)$	$\text{Mean}(K)$	$\text{SD}(K)$	# Overlaps
Ellipsoid 1	21294	(1, 1, 1.4)	0.7125	0.0220	2.5205	0.5254	0
Ellipsoid 2	21294	(1, 1, 1.5)	0.1968	0.0174	1.5927	0.3758	0
Igea	19539	(1, 1, 1.4)	2.6446	0.1005	1.8369	1.0277	0
Hippocampus	43833	(1, 1, 2)	0.1315	0.0440	3.8117	5.8654	0
Rabbit	23954	(1, 1, 1.4)	0.0303	0.0176	3.2448	7.3817	0

of the final density is significantly reduced, which indicates our proposed method effectively equalizes the density. Besides, the mean and standard deviation of K are both small, indicating that the local shape distortions are low in the parameterization results. Moreover, from the number of overlaps, we can see that the mappings are all bijective.

It is natural to ask how the choices of the radii (a, b, c) in the energy (9) will affect the mapping results in the proposed prescribed ellipsoidal method. Here, we consider the Igea model for volume-preserving quasi-conformal parameterizations. Specifically, we use different values of (a, b, c) and analyze the mapping results in terms of volume distortions, local shape error, and bijectivity. In Table 2, we report the mean and standard deviation of volume distortion $\mathcal{D}_{\text{vol}}$, the mean and standard deviation of 3D quasi-conformal dilation K , as well as the number of overlaps. The volume distortion $\mathcal{D}_{\text{vol}}$ for each tetrahedral element T is given by

$$\mathcal{D}_{\text{vol}}(T) = \log \left(\frac{\text{Volume}(f(T)) / \sum_{T' \in \mathcal{T}} \text{Volume}(f(T'))}{\text{Volume}(T) / \sum_{T' \in \mathcal{T}} \text{Volume}(T')} \right). \quad (60)$$

Here, f denotes the solid ellipsoidal parameterization and $\mathcal{T}$ is the set of tetrahedral elements. The above normalization factors and denominator ensure that a perfectly volume-preserving result would yield $\mathcal{D}_{\text{vol}} = 0$. From Table 2, we can observe that the mapping results can reduce the volume distortions effectively for all combinations of the elliptic radii. Moreover, the local shape distortion is more sensitive to the elliptic radii. With different radii, the mean value of K varies from 1.7724 to 2.6091, and the standard deviation varies from 0.7870 to 3.1371. Thus, although the prescribed model effectively redistributes the mass for different ellipsoidal targets, the choice of target radii may have a notable influence on the geometric quality of the map. The number of overlaps is zero for all test examples, demonstrating the effectiveness of our geometric correction scheme.

4.2 Volume-normalized adaptive ellipsoidal model

We next evaluate the volume-normalized adaptive ellipsoidal model. Unlike the above method, this model updates both the volumetric mapping and the ellipsoidal radii. As shown in Fig. 4(a), the initial mass distributions are defined based on the y -coordinates, with the input density varying from 0.5 to 10. After applying our method, the final ellipsoid exhibits a more slender shape. Furthermore, the density histograms indicate that our method reduces the density distortions efficiently. Meanwhile, the histogram of $\log(K)$ shows that the resulting mapping effectively controls

Table 2: **The results of the Igea model obtained by our prescribed ellipsoidal algorithm with different elliptic radii.** Here, for different choices of elliptic radii (a, b, c) of the target solid ellipsoidal domain, we record the mean and standard deviations of volume distortion $|\mathcal{D}_{\text{vol}}|$, the mean and standard deviations of 3D quasi-conformal dilation K , and the number of overlaps.

a	b	c	Mean($ \mathcal{D}_{\text{vol}} $)	SD($ \mathcal{D}_{\text{vol}} $)	Mean(K)	SD(K)	# Overlaps
1	1.0	1.0	0.0492	0.0768	1.9176	0.8729	0
	0.8	1.0	0.0500	0.0806	2.1165	1.0539	0
	1.2	1.0	0.0504	0.0754	1.8612	0.8191	0
	1.4	1.0	0.0513	0.0751	1.9021	0.7870	0
	1.0	0.8	0.0505	0.0783	2.1334	0.9019	0
	1.0	1.2	0.0501	0.0775	1.8314	0.9255	0
	1.0	1.4	0.0508	0.0781	1.8369	1.0277	0
	1.0	1.6	0.0526	0.0862	1.9461	3.1371	0
	0.6	1.2	0.0574	0.0893	2.6091	1.5506	0
	0.8	1.2	0.0516	0.0816	2.0803	1.1362	0
	1.2	1.6	0.0516	0.0833	1.7724	1.8767	0
	1.4	1.8	0.0535	0.0847	1.8009	1.1723	0

local shape distortion. Fig. 4(b) shows another example with the initial density varying with the z -coordinate. The resulting ellipsoid is stretched along the z -direction, while the high-density domains expand and the low-density domains shrink. The density and local shape histograms show that the resulting mapping equalizes the density while maintaining low shape distortions.

Next, we consider computing adaptive ellipsoidal volumetric parameterizations for simply connected 3-manifolds. For each example in Fig. 5, we first map the 3-manifold to a solid ellipsoid with a specific mass distribution and arbitrary radii. Then, we apply our volume-normalized adaptive ellipsoidal algorithm to obtain the ellipsoidal volumetric parameterizations. Compared with the initial mappings, we can see that the shapes of the final results are changed significantly. Moreover, the density histograms show that the resulting parameterizations are highly mass-preserving, while the histograms of $\log(K)$ demonstrate that local shape distortions are effectively controlled.

Table 3 provides a more quantitative analysis for our Volume-normalized adaptive ellipsoidal method. We record the initial and final radii, variance of initial and final density, mean and deviation of 3D quasi-conformal dilation, and the number of overlaps. The differences between initial and final radii indicate that our method effectively adjusts the ellipsoidal shape based on the energy formula. The mean and standard deviation of K are small, demonstrating that the adaptive method can effectively reduce the local shape distortion. Besides, the number of overlaps is 0 for all examples, which indicates the mapping results are folding-free.

4.3 Sea-embedded free-boundary model

Then, we test our sea-embedded free-boundary model for computing free-boundary mass-preserving quasi-conformal mappings. Here, the number of layers is set to $L = 3$ for the following examples. In Fig. 6, we define different mass distributions on the solid ellipsoid and prescribe different continuous input density distributions. After applying our free-boundary method to these ellipsoidal examples, it can be observed that the geometry of the resulting mappings changes significantly, and the final domains are no longer ellipsoidal. Moreover, high-density regions are enlarged while low-density regions are shrunk, resulting in equalized final density distributions. Concurrently, the histograms

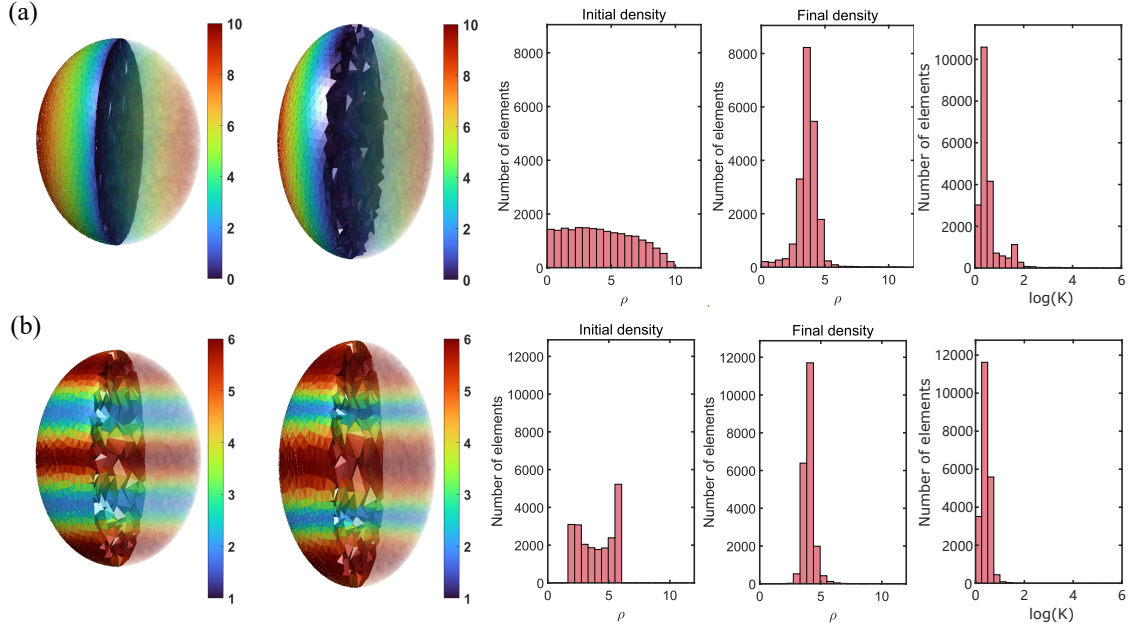


Figure 4: **Volume-normalized adaptive ellipsoidal mass-preserving quasi-conformal maps of solid ellipsoids.** Each row shows one example. (a) An example with a continuous input density varying with the y -coordinate. (b) An example with a continuous input density varying with the z -coordinate. Left to right: The initial solid ellipsoid color-coded with the initial density, the final result color-coded with the initial density, the histogram of the initial density, the histogram of the final density, and the histogram of $\log(K)$.

Table 3: **The performance of the volume-normalized adaptive ellipsoidal model.** For each example, we record the number of tetrahedral elements, the initial radii (a_0, b_0, c_0) , the final radii (a_1, b_1, c_1) , the variance of the normalized initial density $\tilde{\rho}_0 = \frac{\rho_0}{\text{Mean}(\rho_0)}$, the normalized final density $\tilde{\rho}_1 = \frac{\rho_1}{\text{Mean}(\rho_1)}$, the mean and standard deviations of the 3D quasi-conformal dilation K , and the number of overlaps.

Example	# Elements	Initial radii (a_0, b_0, c_0)	Final radii (a_1, b_1, c_1)	$\text{Var}(\rho_0)$	$\text{Var}(\rho_1)$	$\text{Mean}(K)$	$\text{SD}(K)$	# Overlaps
Ellipsoid 3	21294	(1, 1, 1.5)	(1.1395, 0.9319, 1.4126)	0.3684	0.0915	2.3578	8.3280	0
Ellipsoid 4	21294	(1, 0.6, 1.2)	(0.9418, 0.5651, 1.3530)	0.1272	0.0162	1.5275	0.3219	0
Igea	19539	(1, 1, 1)	(0.8891, 1.1248, 1.0000)	2.6008	0.0637	1.7976	0.8411	0
Skull	41938	(1, 1, 1.1)	(0.9412, 1.1008, 1.0611)	0.1638	0.0066	2.1235	0.7820	0
Rabbit	23954	(1, 1, 1.4)	(0.8708, 0.8708, 1.8462)	0.0303	0.0080	3.2706	6.4489	0

of $\log(K)$ demonstrate that local shape distortions remain well-controlled.

In Fig. 7, we present the free-boundary volumetric parameterizations for some simply connected 3-manifolds. In all examples, the proposed method yields mass-preserving quasi-conformal mappings. Specifically, the shape of the target domain is optimized effectively during the iteration process

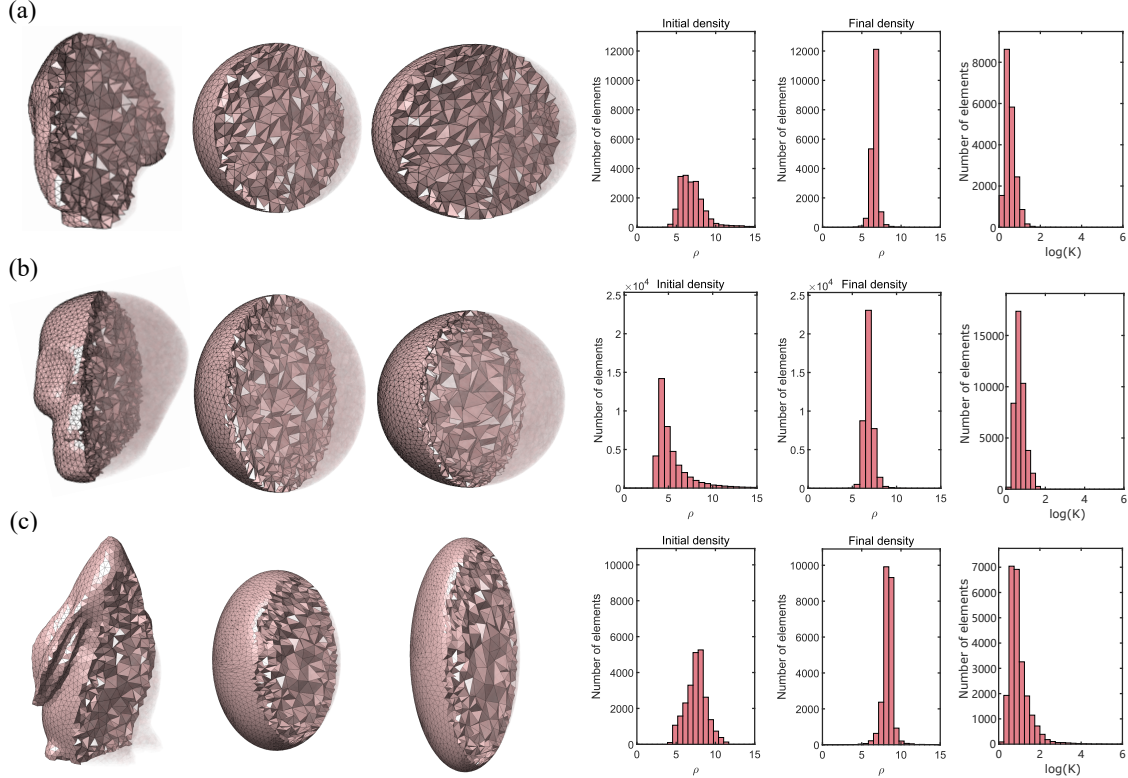


Figure 5: **Volume-normalized adaptive ellipsoidal volumetric parameterization of simply connected 3-manifolds obtained by our method.** Each row shows one example. (a) The Igea model. (b) The Skull model. (c) The Rabbit model. Left to right: The original 3-manifold, the initial solid ellipsoid, the final result, the histogram of the initial density, the histogram of the final density, and the histogram of $\log(K)$.

under the free-boundary condition. Compared with the initial density, the final density becomes highly equalized. Moreover, the local shape distortions are optimized due to the shape change. Table 4 summarizes the initial radii, variance of normalized initial and final densities, mean and standard deviation of K , and the number of overlaps. We can see that the proposed free-boundary method can reduce density distortion significantly while maintaining low local shape distortion. This demonstrates that our algorithm is capable of computing free-boundary mass-preserving quasi-conformal parameterizations for a wide range of simply connected 3-manifolds.

4.4 Comparison of the proposed models and prior solid ball parameterization

For a more quantitative analysis of our parameterization framework, in Table 5 we record the variance of the final density, the mean and standard deviation of K , and the number of overlaps for different target domain settings. For comparison, we also include the solid ball parameterization results [43]

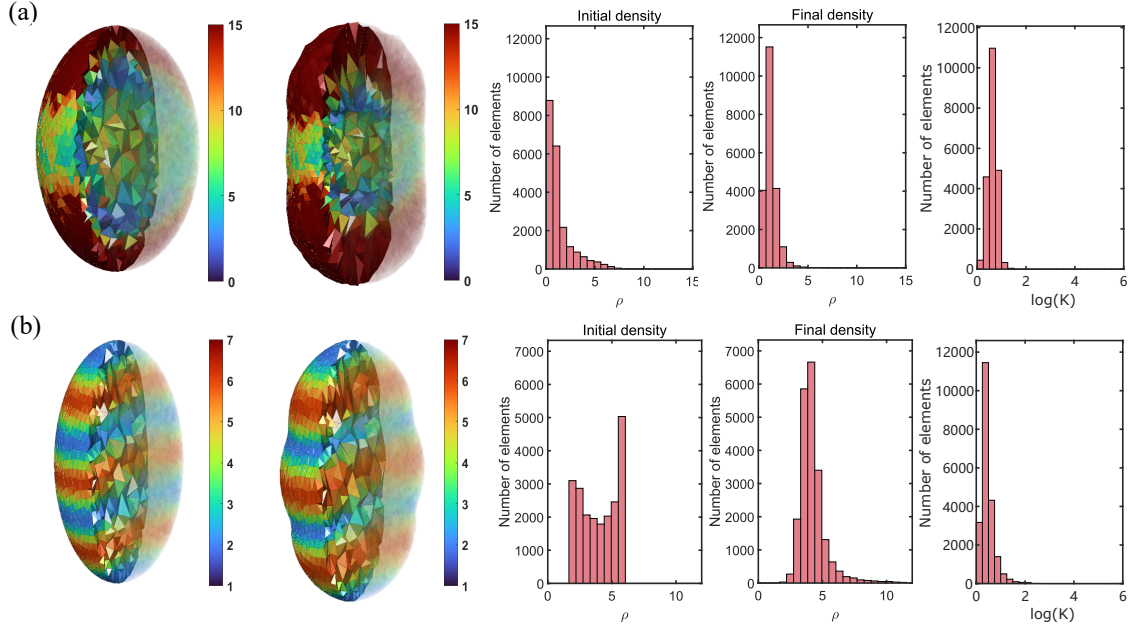


Figure 6: **Free-boundary density-equalizing quasi-conformal maps of ellipsoidal 3-manifolds.** Each row shows one example. (a) An example with a continuous input density varying with the distance to the origin. (b) An example with a continuous input density varying with the z -coordinate. Left to right: The initial solid ellipsoid color-coded with the initial density, the final result color-coded with the initial density, the histogram of the initial density, the histogram of the final density, and the histogram of $\log(K)$.

Table 4: **The performance of sea-embedded free-boundary model.** For each example, we record the number of tetrahedral elements, the initial radii (a, b, c) , the variance of the normalized initial density $\tilde{\rho}_0 = \frac{\rho_0}{\text{Mean}(\rho_0)}$, the normalized final density $\tilde{\rho}_1 = \frac{\rho_1}{\text{Mean}(\rho_1)}$, the mean and standard deviations of the 3D quasi-conformal dilation K , and the number of overlaps.

Example	# Elements	Initial (a, b, c)	$\text{Var}(\rho_0)$	$\text{Var}(\rho_1)$	$\text{Mean}(K)$	$\text{SD}(K)$	# Overlaps
Ellipsoid 5	21294	(1, 1, 1.5)	1.0162	0.3161	1.9055	0.3565	0
Ellipsoid 6	21294	(1, 0.6, 1.6)	0.1242	0.0873	1.6629	0.7046	0
Multi-lobed ball	38917	(1, 1, 1)	0.1438	0.0091	1.8377	2.8455	0
Skull	41938	(1, 1, 1.6)	0.0422	0.0020	2.2285	0.6750	0
Hippocampus	43833	(1, 1, 2)	0.1315	0.0173	3.0606	2.8963	0

as a baseline. Note that for the case of a prescribed ellipsoid, the parameterization performance depends on the choice of the radii and hence is notably different from the solid ball case. Moreover, compared with the solid ball results, the adaptive ellipsoidal and free-boundary models reduce both density and local shape distortions across all examples. Among the three proposed models, the results demonstrate that relaxing the target-domain constraint generally provides additional

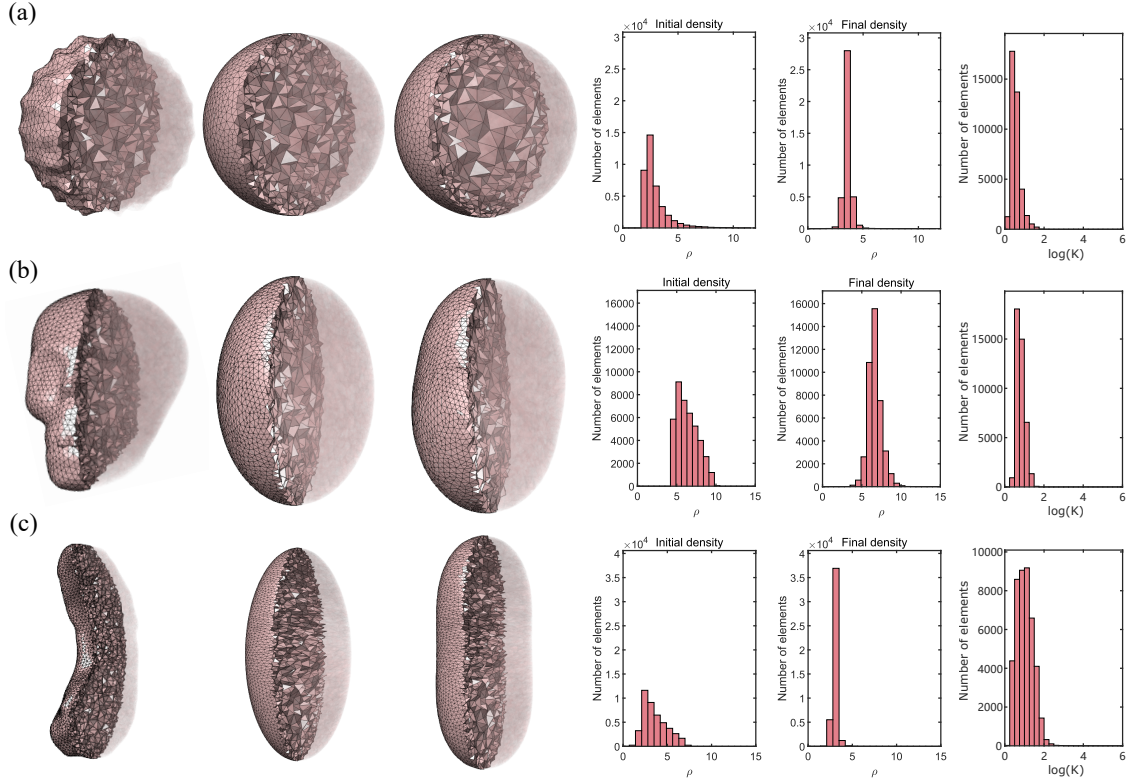


Figure 7: **Free-boundary volumetric parameterization of simply connected 3-manifolds obtained by our method.** Each row shows one example. (a) The Multi-lobed ball model. (b) The Skull model. (c) The Hippocampus model. Left to right: The original 3-manifold, the initial solid ellipsoid, the final result, the histogram of the initial density, the histogram of the final density, and the histogram of $\log(K)$.

flexibility for balancing mass and local shape distortions. In particular, the adaptive ellipsoidal and free-boundary methods consistently reduce the final density variance compared with the prescribed ellipsoidal method. The free-boundary method also substantially reduces the standard deviation of the quasi-conformal dilation K in all three examples, which indicates this method controls local shape distortions effectively. Moreover, the number of overlaps is zero, indicating that all result mappings are folding-free. This comparison highlights the flexibility of our proposed framework for handling 3-manifolds with different geometries.

5 Applications

In this section, we showcase the applicability of our proposed volumetric parameterization framework for 3-manifold remeshing, registration, and morphing.

Table 5: Comparison of the three proposed models and the solid ball parameterization [43] for the Vase, Igea, and David models. For each model, we record the variance of the normalized final density $\hat{\rho}_1 = \frac{\rho_1}{\text{Mean}(\rho_1)}$, mean and standard deviation of K , and the number of overlaps. The chosen/optimized radii for each model are also presented. Note that for the adaptive and free-boundary models, we use the radii in the prescribed case as the initial guess.

Example	Model	$\text{Var}(\hat{\rho}_1)$	$\text{Mean}(K)$	$\text{SD}(K)$	# Overlaps
Vase	Unit ball [43] (1, 1, 1)	0.8855	2.4022	2.6903	0
	Prescribed ellipsoid (1, 1, 1.2)	0.7321	2.3366	2.9153	0
	Adaptive ellipsoid (1, 0.9958, 1.2050)	0.6108	2.3244	2.2324	0
	Free-boundary	0.0705	1.9849	0.7821	0
Igea	Unit ball [43] (1, 1, 1)	0.0480	1.9377	0.8633	0
	Prescribed ellipsoid (1, 1.2, 1)	0.0511	1.8795	0.8053	0
	Adaptive ellipsoid (0.8620, 1.1610, 1.1991)	0.0458	1.6876	0.7971	0
	Free-boundary	0.0286	1.7504	0.5063	0
David	Unit ball [43] (1, 1, 1)	0.7727	2.3921	2.3228	0
	Prescribed ellipsoid (1, 1, 1.4)	0.8781	2.1979	2.2543	0
	Adaptive ellipsoid (0.9537, 1.0129, 1.4493)	0.6537	2.1827	2.0505	0
	Free-boundary	0.5802	2.1087	1.5561	0

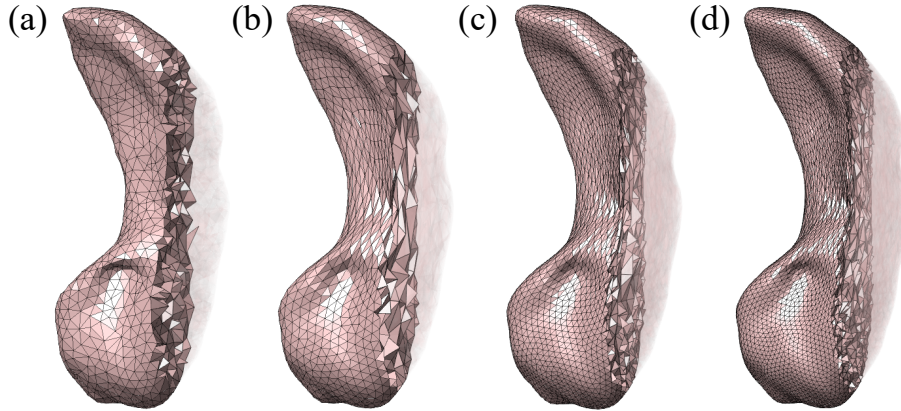


Figure 8: **Multiresolution volumetric remeshing of the Hippocampus model.** (a) The original Hippocampus model. (b)–(d) The remeshed results with approximately 4K, 8K, and 15K vertices, respectively.

5.1 3-manifold Remeshing

Our proposed framework can be easily applied to 3-manifold remeshing. Specifically, given a 3D simply connected manifold $\mathcal{M}$ in $\mathbb{R}^3$, we first compute an ellipsoidal volumetric parameterization $f : \mathcal{M} \rightarrow \mathbb{E}_{a,b,c}$. Then, we can easily generate a uniform tetrahedral mesh in the ellipsoidal domain using existing tools such as DistMesh [51]. Finally, utilizing the inverse mapping f^{-1} , we can map the uniform mesh onto $\mathcal{M}$, yielding a remeshed 3-manifold.

In Fig. 8, we present several examples of remeshing of the brain hippocampus model with different numbers of vertices. Compared with the original model, the remeshed manifolds exhibit high mesh quality, with uniform and regular tetrahedral elements. More specifically, as the resolution increases, the features of the remeshed hippocampus become clearer, especially in the high-curvature domain.

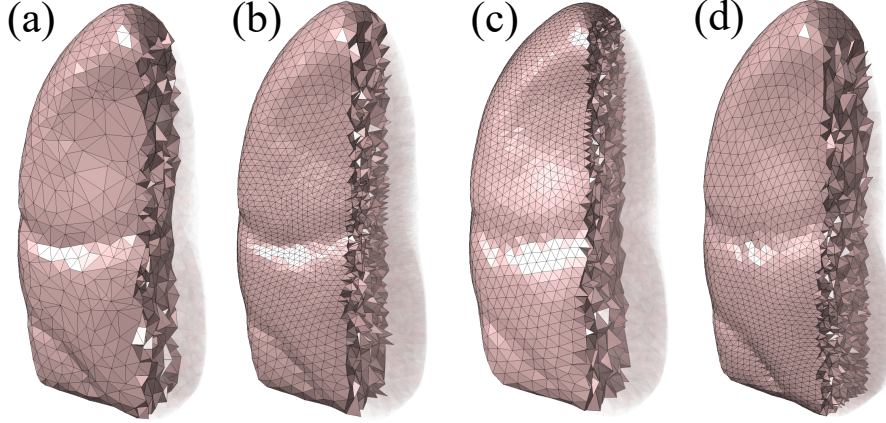


Figure 9: **Localized adaptive volumetric remeshing of the Lung model.** (a) The original Lung model. (b)–(d) show the adaptive mesh reconstruction results after local mesh refinement in the central, top, and bottom regions of the lung, respectively, simulating different possible locations of lung lesions.

Note that the effect of the proposed methods is closely related to the prescribed mass distribution. Specifically, the high-density domain would be enlarged while the low-density domain would be shrunk. Therefore, we can obtain adaptive remeshing results by setting different mass distributions. Fig. 9 shows several adaptive remeshing results of the Lung model to demonstrate this localized adaptivity. Here we consider a Lung model and simulate possible lesions located in different anatomical regions. Specifically, we set prescribed mass distributions concentrated in the central, top, and bottom regions. The original Lung model is shown in Fig. 9(a). Fig. 9(b)–(d) present the remeshing results with different user-defined mass distributions. It can be observed that the mesh density in these special regions is higher than in the other parts.

5.2 3-manifold Registration and Morphing

The proposed volumetric parameterization provides a common canonical domain for establishing volumetric correspondences between simply connected 3-manifolds. Let $\mathcal{M}_0$ and $\mathcal{M}_1$ be two 3-manifolds. Using the prescribed ellipsoidal algorithm, we can compute the volumetric parameterizations $f_0 : \mathcal{M}_0 \rightarrow \mathbb{E}_{a,b,c}$ and $f_1 : \mathcal{M}_1 \rightarrow \mathbb{E}_{a,b,c}$ with fixed radii a, b, c . Here, we set $a = b$ in the solid ellipsoidal domain $\mathbb{E}_{a,b,c}$.

Since the ellipsoid is rotationally symmetric about the principal z -axis, we can remove the rotational degree of freedom by solving

$$\theta_i = \operatorname{argmin}_{\theta \in [0, 2\pi)} \int_{\mathcal{M}_i} \|R_z(\theta)f_i(x) - \widehat{\iota}_i(x)\|_2^2, \quad i = 0, 1, \quad (61)$$

where $\widehat{\iota}_i$ denotes the centered and scale-normalized embedding of $\mathcal{M}_i$ and R_z is the rotation function along z -axis. Then, we define $h = R_z(-\theta_1)R_z(\theta_0) = R_z(\theta_0 - \theta_1)$. Consequently, the ellipsoidal volumetric registration map from $\mathcal{M}_0$ to $\mathcal{M}_1$ is defined as $\phi = f_1^{-1} \circ h \circ f_0$. The illustration of the registration is given in Fig. 10.

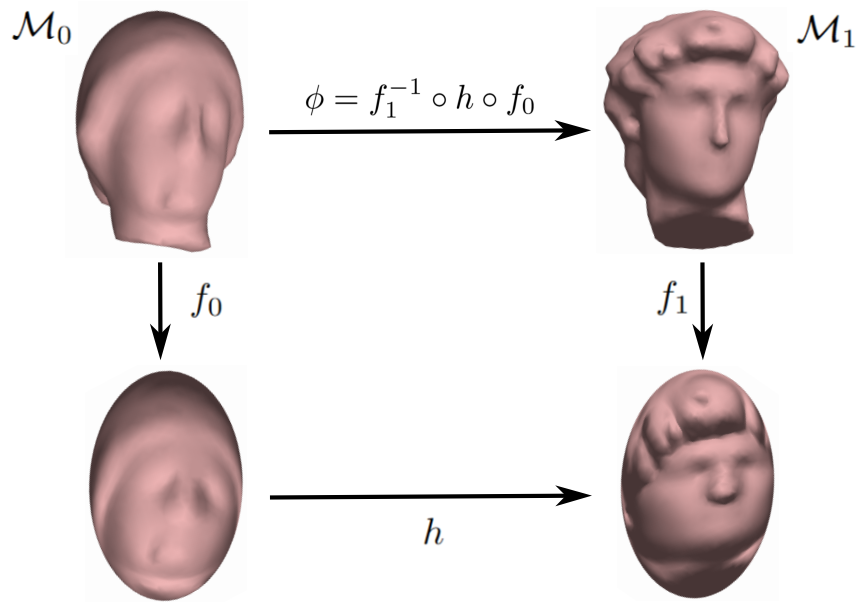


Figure 10: **Registration between simply connected 3-manifolds via our proposed volumetric parameterization methods.** The source and target volumetric meshes $\mathcal{M}_0$ and $\mathcal{M}_1$ are mapped onto the same prescribed solid ellipsoid by the parameterizations f_0 and f_1 , respectively. After aligning their principal z -axes, the remaining rotational freedom is removed by an optimized rotation. The resulting volumetric registration map is given by $\phi = f_1^{-1} \circ h \circ f_0$.

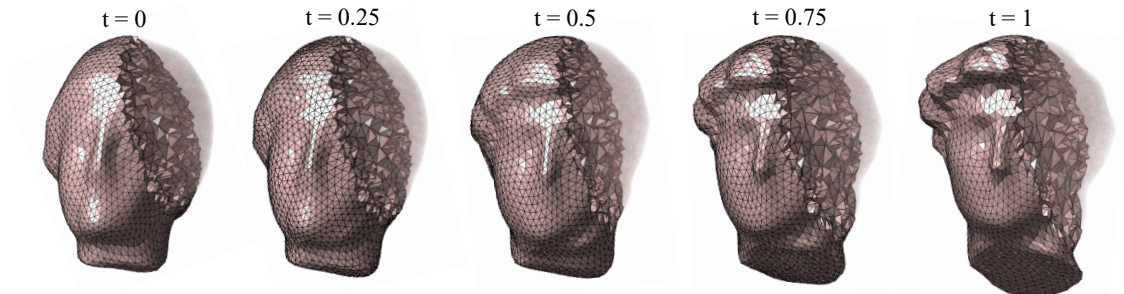


Figure 11: **3-manifold homotopy via our proposed volumetric parameterization methods.** Here, we construct a continuous deformation from the Igea model ($t = 0$) to the David model ($t = 1$). The intermediate volumetric mappings at $t = 0, 0.25, 0.5, 0.75$, and 1 illustrate the continuous transition between the two models. The sectional view shows the internal tetrahedral mesh.

Based on the obtained volumetric registration, we can further construct a continuous morphing sequence between $\mathcal{M}_0$ and $\mathcal{M}_1$. Specifically, the linear homotopy is defined as

$$\mathcal{L}(x, t) = (1 - t)x + t\phi(x). \quad (62)$$

It is clear that $\mathcal{L}(\mathcal{M}_0, 0) = \mathcal{M}_0$ and $\mathcal{L}(\mathcal{M}_0, 1) = \mathcal{M}_1$. In Fig. 11, we present the morphing results between two volumetric meshes at $t = 0, 0.25, 0.5, 0.75$, and 1. The intermediate maps exhibit a smooth and gradual transition in both the overall shape and the local geometric features. Altogether, the above examples demonstrate the applicability of our method for the registration and morphing of volumetric meshes.

6 Conclusion

In this work, we have proposed a novel framework for computing adaptive volumetric parameterization for simply connected 3-manifolds. Our framework simultaneously controls the local shape and mass distortions while optimizing the parametric domain. Depending on the target domain family, we develop three progressively more flexible models: a prescribed ellipsoidal model, a volume-normalized adaptive ellipsoidal model, and a sea-embedded free-boundary model. These models provide increasing flexibility in target geometry, from fixed ellipsoids to adaptively optimized ellipsoids, and even general non-ellipsoidal target domains, thereby allowing us to easily achieve different volumetric mapping effects with minimal geometric distortion. The experimental results have demonstrated the effectiveness of our proposed methods. In particular, our methods can be easily utilized for a large variety of structures ranging from digital models commonly used in computer graphics to medical shapes such as the skull, hippocampus, and lung. Altogether, our proposed methods provide a new approach for the representation and processing of simply connected 3-manifolds for practical applications in manifold remeshing, registration, and morphing.

Note that our proposed volumetric parameterization framework is currently limited to simply connected 3-manifolds. In the future, we plan to extend the methods for manifolds with more complex topologies. Besides, the current methods focus on volumetric parameterization with controllable local shape and mass distortions. A possible future direction is to incorporate landmark-matching constraints into our framework, which can be used for landmark-based 3-manifold registration. Finally, as our current methods are only applicable to tetrahedral meshes, another natural next step is to extend them for point clouds and hexahedral meshes.

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